Digital Forensic Examination Report Apple iPhone 8 — Full iOS Filesystem Analysis

Case Reference: UFDR_CASE_001

Evidence File: Apple iOS Full File system_2021-10-01_Report.ufdr
*SHA-256:*d4ee0bedb24e07653d11ba1eb14f68cd4cb7862e69004a974c2819fce071b523

Examiner: **Alister A. Rodrigues**

Tools: Cellebrite Reader 7.48.1.3 · DB Browser for SQLite 3.12.2

*Examination Date:*October 2025

Examiner Profile

Alister A. Rodrigues is a digital forensics examiner with training in mobile device acquisition, iOS filesystem analysis, and artifact recovery. This examination demonstrates proficiency with Cellebrite Reader for UFDR parsing and DB Browser for SQLite for direct database querying of iOS system databases including sms.db, applicationState.db, and knowledgeC.db.

Executive Summary

This digital forensic investigation involved the analysis of an extracted iOS image file, “Apple iOS Full File system_2021-10-01_Report.ufdr,” provided for examination using Cellebrite Reader version 7.48.1.3. The purpose of the investigation was to identify the characteristics, ownership, network activity, application usage, and location data associated with the device in order to construct a profile of the user’s activity and determine the evidentiary value of recovered artifacts.

A verified working copy of the UFDR file was examined within a controlled forensic environment using Cellebrite Reader and DB Browser for SQLite. Hash verification was performed to ensure data integrity, and all investigative steps followed established best practices as recommended by the National Institute of Standards and Technology (NIST).

The examined image originated from an Apple iPhone 8 (Model D20AP) operating in the (UTC–05:00) America/New York time zone and registered to the Apple ID “bergcybersectesting@gmail.com.” The device exhibited active synchronization with iCloud and contained evidence of continuous use between September 11 and October 1, 2021.

Although no SMS, iMessage, or third-party chat data was recovered, substantial evidence of communication activity was documented through email records, Bluetooth/Wi-Fi logs, and application usage telemetry. The user engaged primarily with health and menstrual tracking applications (MenopauseDiary2, Lunanueva, Menolife, and others) and demonstrated repeated connectivity to local Wi-Fi networks and Bluetooth devices.

Geolocation artifacts totaling over 5,700 entries were recovered, corroborating continuous device movement logging across multiple days in a defined area near 40.73°N, 73.87°W. These findings collectively indicate that the device was actively used by a single individual for personal activities, health tracking, and network communications during the period under review.

Methodology

1.1 Introduction to Methodology

This section outlines the procedures, tools, and verification methods used to examine the digital evidence. Each step was performed in accordance with accepted forensic practices to maintain integrity and ensure results can be reproduced by another qualified examiner.

1.2 Evidence Handling and Integrity Verification

The evidence consisted of one file titled “Apple iOS Full File system_2021-10-01_Report.ufdr.” A UFDR (Universal Forensic Data Report) is a Cellebrite format that stores extracted device data and supporting artifacts.

A verified working copy of the file was created for analysis, while the original remained unchanged. File integrity was confirmed using the SHA-256 hashing algorithm with the following command: certutil -hashfile "Apple iOS Full File system_2021-10-01_Report.ufdr" SHA256

Resulting hash:
d4ee0bedb24e07653d11ba1eb14f68cd4cb7862e69004a974c2819fce071b523

This unique value serves as the file’s digital fingerprint, confirming authenticity throughout the investigation. All evidence was stored in a controlled, isolated forensic working directory C:\Users\Karavense\Desktop\CelleBrite\CaseFiles\UFDR_CASE_001\, with the original file retained as read-only throughout the examination.

In addition to the manual SHA-256 verification conducted using Windows CertUtil, the file hash was also verified by the Cellebrite Reader independently, confirming that the evidence file matched the original extraction without modification.

1.3 Tools and Environment

Analysis was conducted with Cellebrite Reader version 7.48.1.3 on Windows 11 Home (ARM build) in a VMware Fusion virtual machine with Internet access for network artifact lookups. Cellebrite Reader was chosen because it natively parses .ufdr files and organizes extracted data into categories such as device information, contacts, communications, media, and applications.

All actions occurred on the working copy; the original file remained read-only.

DB Browser for SQLite (version 3.12.2) was also utilized to execute structured queries on databases such as sms.db, applicationState.db, and knowledgeC.db, enabling detailed retrieval of app usage, installation events, and device behavior data.

1.4 Time Zone Normalization

When the report opened, Cellebrite Reader detected the device's time zone as (UTC-05:00) America/New York. The option selected was "Yes" to adjust timestamps accordingly, ensuring that message and location times reflected the device's actual local time. All times in this report are shown in that time zone.

1.5 Wireless Network Artifacts

Cellebrite Reader identified Wi-Fi access points (BSSIDs) and cell-tower data capable of location conversion. The notification was acknowledged with "Don't show again," leaving automatic lookups active through the system's Internet connection. This allowed approximate mapping of network and location data during analysis.

1.6 Examination Workflow

After confirming integrity, the report was opened in Cellebrite Reader, which populated standard categories including Device Info, Contacts, Applications, Communications, Media, Wi-Fi, Bluetooth, and Timeline.

The analysis followed these steps:

- Review device metadata (model, serial number, OS version).
- Identify ownership indicators from contacts and accounts.
- Examine call logs, text, and application messages.
- Review installed applications and related activity.
- Analyze timeline, Wi-Fi, Bluetooth, and GPS data for movement patterns.
- Inspect photographs and videos for embedded metadata.
- Export relevant data tables and screenshots for documentation.

Each artifact was examined in read-only mode, and screenshots were labeled and referenced as figures.

1.7 Scope

This investigation examined the full mobile device extraction to determine:

- Device characteristics and ownership
- Personal identifiers (contacts, accounts, emails)

- Geographic movements and wireless connections
- Communication history and application use

All findings are factual observations derived directly from the UFDR data and documented using industry-recognized forensic methods.

Findings

2.1 Device Identification

A Preliminary Device Report was generated directly from Cellebrite Reader to capture the verified hardware and system identifiers of the examined mobile device. The software automatically confirmed the authenticity of the extraction, displaying a verification notice stating "The report's signature is verified." This confirms that the evidence file was produced by Cellebrite's controlled extraction process and has not been modified since acquisition.

The analyzed device was identified as an Apple iPhone 8, hardware model D20AP, operating in the (UTC-05:00) America/New York time zone. The device name, "Berg's iPhone," appeared in both the extraction summary and the generated PDF report. Cellebrite Reader also recorded that Location Services were enabled and that the device maintained synchronization with Apple iCloud, indicating active use by a registered user at the time of the last backup.

Key identifiers extracted from the device are summarized below:

Parameter	Value
Device Name	Berg's iPhone
Detected Phone Model	iPhone 8
Model Number	D20AP
Serial Number	FFMWR1NRC6F
IMEI	35670108761405
Apple ID	bergcybersectesting@gmail.com
iCloud Account Present	True
Last iCloud Backup	2021-09-22 07:41:00 (UTC-05:00)
Time Zone	(UTC-05:00) America/New York

Location Services Enabled	True
Wi-Fi MAC Address	3C:2E:F9:3F:A3:BF
Device Unique ID	27 1b 26 d2 e4 ab 92 f6 0e 2d ed 6c fd b6 e0 97 f0 f9 2d 1

Table 2.1.1 *Preliminary Device Information*

These attributes confirm that the dataset originated from a single physical Apple device configured for personal use. The combination of IMEI, serial number, and Apple ID directly links the extraction to a specific iPhone unit and account, satisfying standard identification requirements under the National Institute of Standards and Technology (NIST) guidelines for mobile evidence documentation.

The presence of active synchronization to iCloud and valid network identifiers (such as the registered Wi-Fi MAC address) indicates continuous participation in Apple's cloud ecosystem. Because Location Services were enabled, the device recorded periodic geolocation data that later contributed to reconstructing movement patterns.

Furthermore, the consistent New York time-zone configuration across both device metadata and extracted artifacts helps verify temporal accuracy when correlating email activity, network connections, and location events discussed in subsequent sections.

Collectively, these findings confirm that the analyzed dataset represents a complete and verified logical extraction from an Apple iPhone 8 registered to a single, identifiable user.

Preliminary Device Report - Apple iOS Full File system

Cellebrite
www.cellebrite.com

Device Information

Name	Value
Legacy	
Phone Settings	
Message Retention Duration	Forever
Location Services Enabled	True
Time Zone	(UTC-05:00) New_York (America)
Model number	D20AP
Unique ID	271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1
Serial	FFMVR1NRJC6F
IMEI	356701087651405
AirDrop ID	1379d67cd290
Last Cloud Backup Date	2021-09-22T20:47:10.000+00:00
Apple ID	bergcybersectesting@gmail.com
iCloud account present	True
Advertising Id (IDFA) #1	9D2136BD-1E9A-4EC6-8109-7D9A3978A63F
Device Name	Berg's iPhone
Detected Phone Model	iPhone 8
Sync Data	
VoiceMemo Size (bytes)	0
VoiceMemo Entries	0
UserData Size (bytes)	705683456
UserData Entries	3225
Storage available (Bytes)	55224352768
Book Size (bytes)	0
Book Entries	0
Application Size (bytes)	0
Application Entries	0
Logs Size (bytes)	2449408
Logs Entries	80
Ringtone Size (bytes)	0
Ringtone Entries	0
Proofing Size (bytes)	0
Proofing Entries	0
Storage capacity (Bytes)	57333288960
Sync host name	Computer: XVXD4SH\User: alexo
Network Interfaces	
Wi-Fi MAC address	3C:2E:F9:3F:A3:BF
MAC address	3E:2E:F9:3F:A3:C0
MAC address	3E:2E:F9:3F:A3:3F

Figure 2.1.1: *Excerpt from the Cellebrite-generated Preliminary Device Report showing verified device metadata for "Berg's iPhone 8."*

2.2 Owner Identification

The extracted dataset contained user account metadata indicating a single consistent ownership profile. Cellebrite's report listed a total of 13 user account records, most of which were linked to the same Apple ID and Gmail address: bergcybersectesting@gmail.com.

Each account entry was parsed from the /private/var/mobile/Library/Accounts/Accounts3.sqlite database, which stores credentials and service associations for active iOS user accounts. The records spanned multiple Apple services including iTunes Store, iCloud, Find My Friends, Game Center, and Messages, all created within

a short timeframe on September 11, 2021, with one additional record dated September 19, 2021, for Gmail. This grouping strongly indicates a single initial configuration or device restoration event.

Collectively, the account data suggests that the device belonged to or was primarily used by an individual associated with the name “Berg Cybersec” and the email address bergcybersectesting@gmail.com. Multiple linked Apple services confirm that this was an active, personalized device rather than a test or factory state.

The synchronization of account creation timestamps across Apple services indicates that the accounts were configured simultaneously, likely during the phone’s initial setup or an iCloud-based restoration. Additionally, the presence of Find My Friends and Device Locator accounts implies that the user enabled Apple’s built-in location-tracking features, reinforcing continuity between this account and the device movements later detailed in Section 2.5 (Device Location Findings).

The inclusion of both local and cloud-linked accounts indicates that the user accessed Apple services both online and offline, maintaining cached credentials for persistent sign-ins. No evidence of additional user profiles or secondary Apple IDs was found within the dataset.

8	**Creation time** 9/11/2021 11:51:28 PM(UTC-4) **Username:** bergcybersectesting@gmail.com **Service Type:** 1291E2D8-AE0A-47F7-BAF1-DA480DB6485B **Source:** Accounts			Account Description: Apple ID **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Accounts/Accounts3.sqlite-wal : 0x1845AF (Table: ZACCOUNT, Size: 1606832 bytes) 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Accounts/Accounts3.sqlite : 0x252D5 (Size: 229376 bytes)	
9	**Creation time** 9/11/2021 11:51:36 PM(UTC-4) **Username:** bergcybersectesting@gmail.com **Service Type:** 7FFAABDB-A9A7-4FC6-97BA-CB4D1F14352F **Source:** Accounts			Account Description: Game Center **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Accounts/Accounts3.sqlite-wal : 0x18533A (Table: ZACCOUNT, Size: 1606832 bytes) 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Accounts/Accounts3.sqlite : 0x26F86 (Size: 229376 bytes)	
10	**Creation time** 9/11/2021 11:51:36 PM(UTC-4) **Username:** bergcybersectesting@gmail.com **Service Type:** 73188EE2-8896-412D-9A8D-A9F465BD1EBA **Source:** Accounts			Account Description: Messages **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Accounts/Accounts3.sqlite-wal : 0x1847F6 (Table: ZACCOUNT, Size: 1606832 bytes) 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Accounts/Accounts3.sqlite : 0x26769 (Size: 229376 bytes)	
11	**Creation time** 9/11/2021 11:51:36 PM(UTC-4) **Username:** bergcybersectesting@gmail.com **Service Type:** 2CC3A6BA-D991-4001-8581-422BDF40E5F0 **Source:** Accounts			Account Description: CloudKit **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Accounts/Accounts3.sqlite-wal : 0x184759 (Table: ZACCOUNT, Size: 1606832 bytes) 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Accounts/Accounts3.sqlite : 0x25288 (Size: 229376 bytes)	
12	**Creation time** 9/11/2021 11:51:36 PM(UTC-4) **Username:** bergcybersectesting@gmail.com **Service Type:** BFB2AE7D-613E-47E8-A21D-5590D772DB3A **Source:** Accounts			Account Description: iCloud **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Accounts/Accounts3.sqlite-wal : 0x18489A (Table: ZACCOUNT, Size: 1606832 bytes) 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Accounts/Accounts3.sqlite : 0x26A73 (Size: 229376 bytes)	
13	**Creation time** **Account Name:** Berg Cybersec **Source:** ScreenTime			Age group: Unknown **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Application Support/com.apple.remotemanagementd/RMAdminStore-Local.sqlite-wal : 0x81C2C (Table: ZCOREUSER, Size: 535632 bytes)	

Figure 2.2.1: *Excerpt from Cellebrite's User Accounts report showing multiple Apple-linked accounts registered to bergcybersectesting@gmail.com, confirming single-user ownership.*

2.3 Communications Analysis

2.3.1 Email Communications

Analysis of the Messages → Emails dataset within Cellebrite Reader revealed a single configured email account: bergcybersectesting@gmail.com, hosted on Gmail and actively used between approximately July 2020 and September 2021. The folder structure identified during examination [Gmail]/Sent (3) and INBOX (131), shows that both outbound and inbound communications were retained on the device.

Account Overview

The email artifacts were extracted from the device's mail directories at /private/var/mobile/Library/Mail/ and corresponding SQLite index tables, confirming the data originated from the Apple Mail client configured on the device. Cellebrite Reader automatically calculated MD5 hashes for each .eml file, ensuring message authenticity and confirming that no data modifications occurred during parsing.

Outbound Messages

Three outgoing emails were located under [Gmail]/Sent, all originating from bergcybersectesting@gmail.com and addressed to alexongkowijoyo91@ymail.com.
The emails were transmitted within a short time span on September 18–19, 2021, and contained the following content indicators:

- One message titled "Bro" with minimal text body content, showing the sender signature "Berg CyberSec testing."
- A second email labeled "Sent from my iPhone" containing two attachments (image0.png and image1.png), both successfully extracted.
- A third message with the subject "History of Period Tracker."

These transmissions indicate personal correspondence rather than automated or system notifications, demonstrating that the device owner used this Gmail account for direct interpersonal communication. Each message carried a standard Sent flag, confirming authenticated SMTP activity from the configured mail client.

Inbound Messages

The INBOX (131) folder contained predominantly automated notifications and account-related messages, establishing a behavioral profile of the user's routine activity. Key examples include:

- Security and verification alerts from Facebook, Google, Apple, and WomanEze, showing regular sign-ins, password resets, and account verification events.
- Subscription and service emails from ProtonVPN, Flo (health app), and Apple receipts, reflecting engagement with privacy utilities, mobile applications, and online transactions.

- System alerts from Find My iPhone, confirming that Apple's location-tracking service was temporarily disabled and later re-enabled, an action consistent with deliberate user intervention.

Of particular note, a ProtonVPN registration email dated September 24, 2021, confirmed successful account creation for bergcybersectesting@gmail.com. This suggests the user actively utilized security-oriented services, indicating awareness of privacy or anonymity tools.

Email Activity Summary

The chronological distribution and message types suggest that bergcybersectesting@gmail.com served as the user's primary identity for both personal and service-related communications. No secondary or alternative mailboxes were detected. The balance between inbound notifications and a limited number of sent personal messages reflects typical usage behavior of an individual Gmail account on a mobile device, rather than automated or high-volume mailing activity.

Cellebrite's embedded metadata corroborated message authenticity through consistent timestamps, verified MD5 hashes, and coherent source path records.

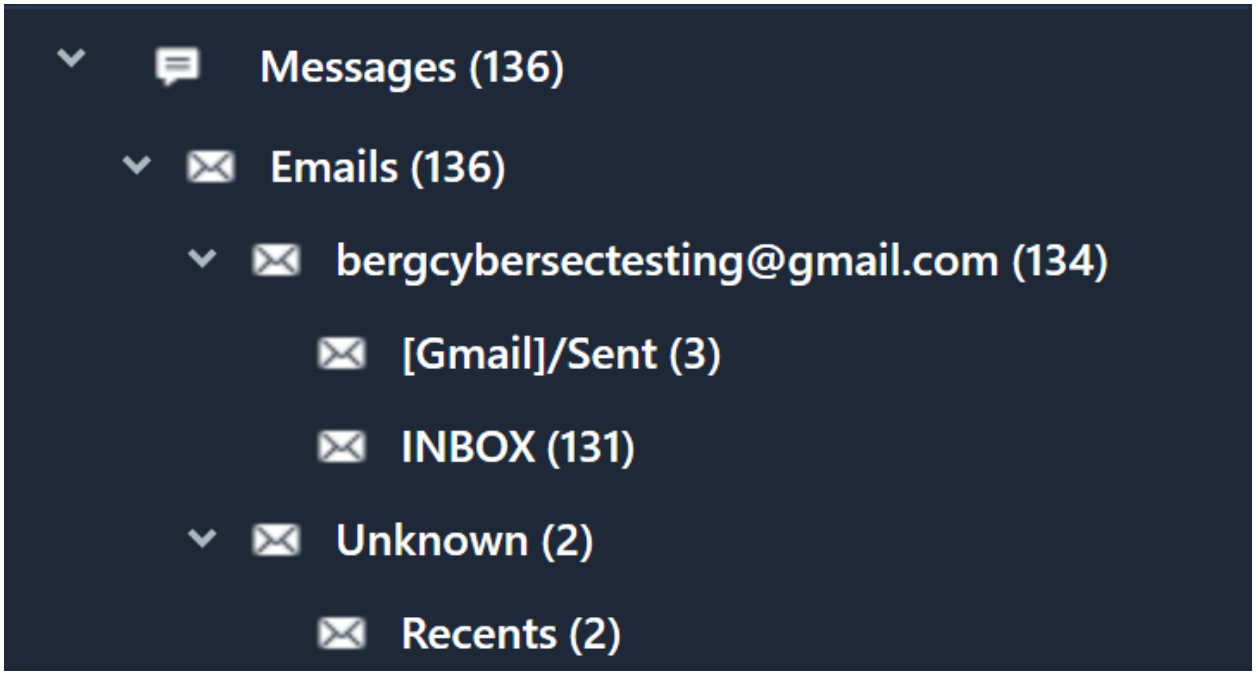

Figure 2.3.1: *Email folder structure for bergcybersectesting@gmail.com showing [Gmail]/Sent and INBOX activity.*

Emails (3)

#	Parties	Content	Other	Deleted
1	**Timestamp:** 9/19/2021 12:24:17 AM(UTC-4) **From:** bergcybersectesting@gmail.com BergCyberSec testing **To:** alexongkowijoyo91@ymail.com	**Subject:** Bro **Body:** **Email header:** **Body file:** mes-1.eml	**Status:** Sent **Account:** bergcybersectesting@gmail.com **Snippet:** **MD5:** f414fd60dfd23dd87636b8f1d11a0c20 **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Mail/Envelope Index : 0x924B2 (Table: messages, mailboxes, Size: 606208 bytes) 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Mail/Protected Index-wal : 0x4ECD1 (Table: addresses, Size: 391432 bytes)	
2	**Timestamp:** 9/19/2021 12:22:01 AM(UTC-4) **From:** bergcybersectesting@gmail.com BergCyberSec testing **To:** alexongkowijoyo91@ymail.com	**Subject:** No subject **Body:** Sent from my iPhone **Email header:** **Attachments:** image0.png image0.png **Body file:** mes-2.eml	**Status:** Sent **Account:** bergcybersectesting@gmail.com **Snippet:** Sent from my iPhone **Priority:** Normal **MD5:** 167a8fbb0757fa20512d4207c3f81a9f **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Mail/Envelope Index : 0x92534 (Table: messages, mailboxes, Size: 606208 bytes) 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Mail/MessageData/131/partial.emlx : 0x0 (Size: 1223 bytes) 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Mail/Protected Index-wal : 0x4C27E (Table: addresses, summaries, Size: 391432 bytes) 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Mail/MessageData/131/Attachments/1/image0.png : (Size: 241381 bytes)	
3	**Timestamp:** 9/18/2021 11:09:28 PM(UTC-4) **From:** bergcybersectesting@gmail.com BergCyberSec testing **To:** alexongkowijoyo91@ymail.com	**Subject:** History of Period Tracker **Body:** **Email header:** **Body file:** mes-3.eml	**Status:** Sent **Account:** bergcybersectesting@gmail.com **Snippet:** **MD5:** ba0cf7adf48226b711d3140915c6492f **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Mail/Envelope Index : 0x925B0 (Table: messages, mailboxes, Size: 606208 bytes) 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Mail/Protected Index-wal : 0x4ECD9 (Table: addresses, Size: 391432 bytes)	

1

Figure 2.3.2: *Excerpt from Cellebrite Extraction Report showing three outgoing emails sent from bergcybersectesting@gmail.com to alexongkowijoyo91@ymail.com on September 18–19, 2021. Two messages contain attached image files, confirming active email use from the device's configured Gmail account.*

Emails (131)

#	Parties	Content	Other	Deleted
1	**Timestamp:** 9/24/2021 1:53:07 PM(UTC-4) **From:** contact@protonvpn.com ProtonVPN **To:** bergcybersectesting@gmail.com **Direction:** Incoming	**Subject:** Welcome to the ProtonVPN community **Body:** Thank you for signing up for ProtonVPN. With ProtonVPN, you can secure your Internet connection with just a few easy steps. [https://protonvpn.com/?utm_campaign=ww-all-2a-vpn-coms_email-g_eng-onboarding&utm_source=proton_users_vpn&utm_medium=email&utm_content=welcome_free] Welcome to the ProtonVPN community! Thank you for signing up for ProtonVPN. With ProtonVPN, you can secure your internet connection with just a few easy steps. [https://protonvpn.com/download?utm_campaign=ww-all-2a-vpn-coms_email-g_eng-onboarding&utm_source=proton_users_vpn&utm_medium=email&utm_content=welcome_free] We believe the internet should be private by default We created ProtonVPN to make a free and uncensored internet available to all. Because we don't make money by collecting or monetizing your data in any way, our service is supported entirely by our community of users who upgrade to paid plans. Our Free plan will always be completely free, with no ads, no logs and no brow **Email header:** Item was truncated due to length **Attachments:** logo.png logo.png (Empty File) devices.png devices.png (Empty File) downloads.png downloads.png (Empty File) facebook.png facebook.png (Empty File) twitter.png twitter.png (Empty File) reddit.png reddit.png (Empty File) link.png link.png (Empty File) **Body file:** mes-1.eml	**Status:** Unread **Account:** bergcybersectesting@gmail.com **Snippet:** Thank you for signing up for ProtonVPN. With ProtonVPN, you can secure your Internet connection with just a few easy steps. Welcome to the ProtonVPN community! Thank you for signing up for ProtonVPN. With ProtonVPN, you can secure your internet connection with just a few easy steps. We believe the internet should be private by default We created ProtonVPN to make a free and uncensored internet available to all. Because we don't make money by collecting or monetizing your data in any way, our ser **Priority:** Normal **MD5:** 9a0e8f667a5e659d0a3ea4f02cd6dcde **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Mail/Envelope Index : 0x93F36 (Table: messages, mailboxes, Size: 606208 bytes) 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Mail/Protected Index-wal : 0x4EBDD (Table: addresses, summaries, Size: 391432 bytes) 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Mail/MessageData/147/partial.emlx : 0x0 (Size: 6179 bytes) 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Mail/MessageData/147/partial.emlx/logo.png : (Size: 0 bytes)	

Figure 2.3.3: *Extract from Cellebrite Email Report displaying an incoming message from ProtonVPN to bergcybersectesting@gmail.com, confirming account registration and associated attachments. This demonstrates the user's engagement with encrypted communication and privacy-oriented services.*

2.3.2 Messaging and Interaction Activity

The forensic examination of the device image included a targeted review for conventional text communications, such as SMS, iMessage, and third-party chat applications (e.g., WhatsApp, Facebook Messenger, Telegram, and Signal). Multiple data sources within the extracted Apple iOS full file system were inspected, including:

- System directories under /private/var/mobile/Library/ and /private/var/db/ containing message databases and CoreDuet logs.
- Cellebrite-parsed categories such as Messages, Application Usage, Contacts, and Device Events.
- InteractionC and system log datasets, which typically record communication metadata when direct content is unavailable.

No recoverable text messages, iMessages, or third-party chat artifacts were identified in the extracted data. The Messages category contained no readable message threads or database remnants, and no populated message databases (such as ChatStorage.sqlite or message.db) were present within the user's data partition.

While InteractionC and CoreDuet artifacts were present, their content reflected only system-level interaction metadata, such as timestamps, application states, and process identifiers, without corresponding user content. Given their limited evidentiary value and lack of correlation with identifiable communications, these datasets were not included as standalone exhibits in this report.

The absence of recoverable message content may be attributed to one or more of the following:

- Data deletion or device restoration prior to acquisition.
- Use of cloud-based or encrypted third-party services not configured for local message retention.
- Extraction limitations preventing full decryption of certain containers under iOS security constraints.

Although no conventional text or chat messages were recovered, cross-referencing of email records, account data, and system telemetry still provided substantial insight into the user's communication behavior and online interactions.

2.4 Network and Connectivity Findings

The extracted device image contained numerous artifacts associated with network and peripheral connectivity. These artifacts were primarily sourced from the following parsed datasets within the Cellebrite Reader report:

- Device Connectivity (180 entries) – predominantly Bluetooth-related artifacts.

- Device Events (463 entries) – including system-level network state changes such as Wi-Fi connections, disconnections, plug-in events, and display activity.
- Wireless Networks (3,609 entries) – representing Wi-Fi access points detected or connected to by the device.

Bluetooth Connectivity

The Device Connectivity dataset revealed multiple Bluetooth connection attempts and detections. Most entries listed device identifiers as "Unknown" with randomized Bluetooth MAC addresses (e.g., F7:16:3C:38:6B:DE, ED:31:1A:02:E2:2F). These randomized identifiers are consistent with Bluetooth privacy features implemented in modern iOS versions, which periodically rotate MAC addresses to prevent persistent device tracking.

Although paired device names were not resolved, the consistent presence of detected Bluetooth signals throughout the extraction timeframe indicates that the iPhone maintained active Bluetooth functionality and operated in an environment containing other discoverable Bluetooth-enabled devices.

Device Connectivity (180)

#	Device Name	Device type	Start time	Connectivity method	Device Identifiers	Additional info	Source	Deleted
1		Unknown		Bluetooth **Connectivity nature:** Detected	**Address:** Random F7:16:3C:3B:68:DE	**Artifact Family:** **Source Repository Path:**	**Source:** **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/containers/Shared/SystemGroup/ACD57B49-6C8D-40AD-AD2A-6B21DB7EBD14/Library/Database/com.apple.MobileBluetooth.ledevices.other.db-wal : 0xF7715 (Table: OtherDevices, Size: 1013552 bytes)	
2		Unknown		Bluetooth **Connectivity nature:** Detected	**Address:** Random ED:31:1A:02:E2:2F	**Artifact Family:** **Source Repository Path:**	**Source:** **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/containers/Shared/SystemGroup/ACD57B49-6C8D-40AD-AD2A-6B21DB7EBD14/Library/Database/com.apple.MobileBluetooth.ledevices.other.db-wal : 0xF76C8 (Table: OtherDevices, Size: 1013552 bytes)	
3		Unknown		Bluetooth **Connectivity nature:** Detected	**Address:** Random F4:60:16:59:EF:A5	**Artifact Family:** **Source Repository Path:**	**Source:** **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/containers/Shared/SystemGroup/ACD57B49-6C8D-40AD-AD2A-6B21DB7EBD14/Library/Database/com.apple.MobileBluetooth.ledevices.other.db-wal : 0x4D73F (Table: OtherDevices, Size: 1013552 bytes)	
4		Unknown		Bluetooth **Connectivity nature:** Detected	**Address:** Random F3:47:B8:78:7D:6A	**Artifact Family:** **Source Repository Path:**	**Source:** **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/containers/Shared/SystemGroup/ACD57B49-6C8D-40AD-AD2A-6B21DB7EBD14/Library/Database/com.apple.MobileBluetooth.ledevices.other.db-wal : 0x4D6F5 (Table: OtherDevices, Size: 1013552 bytes)	

Figure 2.4.1 *Extracted Bluetooth connectivity artifacts showing detected nearby devices with randomized MAC identifiers.*

Wi-Fi Connectivity and Network Activity

The Wireless Networks dataset (3,609 entries) provided a detailed record of Wi-Fi BSSID detections and connections, each entry associated with a timestamp and geolocation coordinate (latitude/longitude). For instance, multiple Wi-Fi access points (e.g., 00:D0:2D:99:29:9F, 00:CB:51:4C:DA:1E, 18:A6:F7:F6:1C:3C) were logged within the same time frame on October 1, 2021, correlating to clustered coordinates around approximately (40.739°, -73.878°). This clustering suggests the device was likely stationary or within a confined geographic area during that period.

This level of Wi-Fi telemetry enables potential location triangulation or environmental reconstruction when cross-referenced with other artifacts such as Device Locations or PowerLog entries.

Wireless Networks (3609)

#	BSSID\SSID	Security Mode	Password	Road Position	Timestamp	Additional info	Source Info	Deleted
1	**BSSID:** 00:D0:2D:99:29:9F			(40.739929, -73.878586)	10/1/2021 2:32:58 AM(UTC-4)	**Package:** **Connection Type:** Unknown **Artifact Family:** **Source Repository Path:**	**Source:** **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/root/Library/Caches/locationd/cache_encryptedB.db : 0x5080AA (Table: WifiLocation, Size: 44048384 bytes)	
2	**BSSID:** 00:CB:51:4C:DA:1E			(40.739189, -73.878319)	10/1/2021 2:32:58 AM(UTC-4)	**Package:** **Connection Type:** Unknown **Artifact Family:** **Source Repository Path:**	**Source:** **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/root/Library/Caches/locationd/cache_encryptedB.db : 0x5080E9 (Table: WifiLocation, Size: 44048384 bytes)	
3	**BSSID:** 00:A0:F8:47:EC:97			(40.739559, -73.878274)	10/1/2021 2:32:58 AM(UTC-4)	**Package:** **Connection Type:** Unknown **Artifact Family:** **Source Repository Path:**	**Source:** **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/root/Library/Caches/locationd/cache_encryptedB.db : 0x508128 (Table: WifiLocation, Size: 44048384 bytes)	
4	**BSSID:** 00:08:30:70:5E:DB			(40.739494, -73.878319)	10/1/2021 2:32:58 AM(UTC-4)	**Package:** **Connection Type:** Unknown **Artifact Family:** **Source Repository Path:**	**Source:** **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/root/Library/Caches/locationd/cache_encryptedB.db : 0x508166 (Table: WifiLocation, Size: 44048384 bytes)	
5	**BSSID:** 00:08:30:70:5E:D7			(40.739490, -73.878312)	10/1/2021 2:32:58 AM(UTC-4)	**Package:** **Connection Type:** Unknown **Artifact Family:** **Source Repository Path:**	**Source:** **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/root/Library/Caches/locationd/cache_encryptedB.db : 0x5081A4 (Table: WifiLocation, Size: 44048384 bytes)	

Figure 2.4.2 *Extracted Wi-Fi network data displaying BSSIDs, coordinates, and timestamps of detected networks.*

Device Event Correlation

Review of the Device Events dataset revealed corresponding Wi-Fi state transitions, including:

- Wi-Fi State: Connect events recorded on September 29 and October 1, 2021.

- Concurrent Display On/Off, Orientation Change, and Device Lock/Unlock events, indicating active user interaction during these connection times.

These correlated events demonstrate that network connectivity occurred while the device was actively in use, rather than passively in standby mode.

202	**Start time:** 9/29/2021 8:34:03 PM(UTC-4) **End time:**	Wifi State	Connect		**Source:** Powerlog **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/containers/Shared/SystemGroup/8F3D52D7-16AB-49C8-8558-57CDC3FE462C/Library/BatteryLife/Archives/powerlog_2021-09-29_ED5AECB2.PLSQL.gz/powerlog_2021-09-29_ED5AECB2.PLSQL : 0x10ECD8 (Size: 5771264 bytes)	
203	**Start time:** 9/29/2021 9:32:13 PM(UTC-4) **End time:**	Wifi State	Connect		**Source:** Powerlog **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/containers/Shared/SystemGroup/8F3D52D7-16AB-49C8-8558-57CDC3FE462C/Library/BatteryLife/CurrentPowerlog.PLSQL : 0x10EFF7 (Size: 6168576 bytes)	
204	**Start time:** 10/1/2021 1:35:16 AM(UTC-4) **End time:** 10/1/2021 1:35:17 AM(UTC-4)	Device Plugin Status	Plugged in		**Source:** KnowledgeC **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db : 0x39379B (Table: ZOBJECT, Size: 4263936 bytes)	
205	**Start time:** 10/1/2021 1:35:16 AM(UTC-4) **End time:** 10/1/2021 1:35:24 AM(UTC-4)	Orientation Change	Orientation landscape		**Source:** KnowledgeC **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db : 0x393701 (Table: ZOBJECT, Size: 4263936 bytes)	
206	**Start time:** 10/1/2021 1:35:16 AM(UTC-4) **End time:** 10/1/2021 1:35:24 AM(UTC-4)	Display On/Off	Display on		**Source:** KnowledgeC **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db : 0x393669 (Table: ZOBJECT, Size: 4263936 bytes)	
207	**Start time:** 10/1/2021 1:35:17 AM(UTC-4) **End time:** 10/1/2021 12:37:43 PM(UTC-4)	Device Plugin Status	Plugged in		**Source:** KnowledgeC **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db : 0x3F17D8 (Table: ZOBJECT, Size: 4263936 bytes)	
208	**Start time:** 10/1/2021 1:35:48 AM(UTC-4) **End time:** 10/1/2021 1:36:00 AM(UTC-4)	Display On/Off	Display on		**Source:** KnowledgeC **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db : 0x393361 (Table: ZOBJECT, Size: 4263936 bytes)	
209	**Start time:** 10/1/2021 3:00:08 AM(UTC-4) **End time:** 10/1/2021 3:00:20 AM(UTC-4)	Display On/Off	Display on		**Source:** KnowledgeC **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db : 0x3AE36C (Table: ZOBJECT, Size:	

Figure 2.4.3 *Extracted device event log showing Wi-Fi connection events and concurrent device activity indicators.*

Although no identifiable personal or enterprise network SSIDs were extracted, the large number of network and connectivity records confirms that the device maintained frequent communication with nearby wireless infrastructure. These artifacts assist in establishing temporal and spatial context, validating user activity windows, and supporting potential correlation with external data sources (e.g., router logs, Wi-Fi access records, and geolocation services).

2.5 Device Location Findings

The device image contained a large volume of geolocation artifacts retrieved from multiple sources within the iOS file system. In total, 5,790 location records were extracted, of which 1,860 entries were identified as deleted but recoverable (Figure 2.5.1). These data points were parsed by Cellebrite Reader primarily from Apple's Core Location and Wi-Fi location caching databases located at /private/var/mobile/Library/Caches/com.apple.routined/ and /private/var/root/Library/Caches/locationd/. The presence of both directories confirms that the dataset included a combination of native GPS and Wi-Fi–derived positioning records.

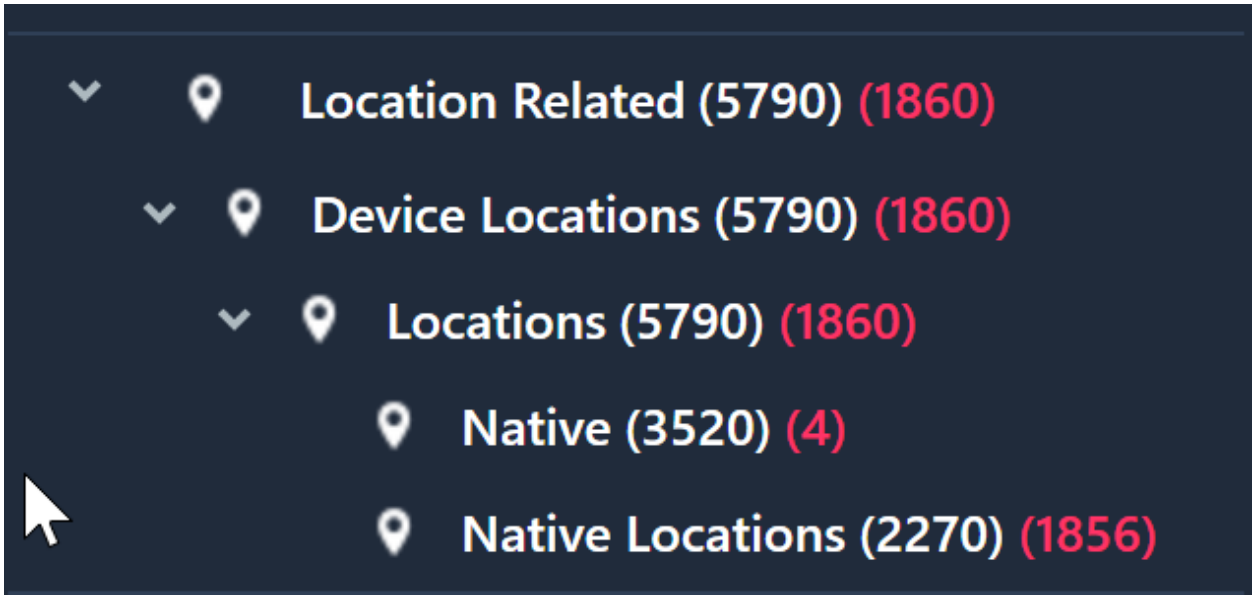

Figure 2.5.1 *Extracted location data summary showing 5,790 total and 1,860 deleted entries from native and Wi-Fi sources.*

Location Data Overview

The extracted GPS dataset contained two distinct types of records:

- Visited locations – recorded during active device use and associated with app- or system-level location requests.

- Harvested locations – captured through background services such as Wi-Fi scanning and Core Location caching, providing continuous passive updates.

Each record included latitude and longitude coordinates, timestamps, precision metrics, and source file references. Coordinates clustered around approximately 40.73°N, 73.87°W, indicating a consistent geographic region of device activity across multiple collection intervals. Temporal metadata showed repeated location logging between September 23 and October 1, 2021, confirming that Location Services were enabled and actively recording during that period.

88		(40.737839, -73.879332)	**Type:** Visited **Precision:** Horizontal: 65 **Time:** 10/1/2021 5:05:30 AM(UTC-4) **End time:** 10/1/2021 5:05:46 AM(UTC-4) **Aggregated locations:** 2 **Source:** Native Locations **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Caches/com.apple.routined/Cache.sqlite-wal : 0x18480B (Table: ZRTCLLOCATIONMO, Size: 3485552 bytes)			
89		(40.737753, -73.879353)	**Type:** Visited **Precision:** Horizontal: 65 **Time:** 10/1/2021 4:35:34 AM(UTC-4) **End time:** 10/1/2021 4:35:34 AM(UTC-4) **Source:** Native Locations **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Caches/com.apple.routined/Cache.sqlite-wal : 0x184873 (Table: ZRTCLLOCATIONMO, Size: 3485552 bytes)			
90		(40.737822, -73.879359)	**Type:** Visited **Precision:** Horizontal: 65 **Time:** 10/1/2021 4:20:10 AM(UTC-4) **End time:** 10/1/2021 4:20:10 AM(UTC-4) **Source:** Native Locations **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Caches/com.apple.routined/Cache.sqlite-wal : 0x1848A7 (Table: ZRTCLLOCATIONMO, Size: 3485552 bytes)			
91		(40.737903, -73.879349)	**Type:** Visited **Precision:** Horizontal: 65 **Time:** 10/1/2021 4:08:32 AM(UTC-4) **End time:** 10/1/2021 4:08:49 AM(UTC-4) **Aggregated locations:** 2 **Source:** Native Locations **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Caches/com.apple.routined/Cache.sqlite-wal : 0x1848DB (Table: ZRTCLLOCATIONMO, Size: 3485552 bytes)			
92		(40.737870, -73.879339)	**Type:** Visited **Precision:** Horizontal: 65 **Time:** 10/1/2021 3:32:51 AM(UTC-4) **End time:** 10/1/2021 3:34:50 AM(UTC-4) **Aggregated locations:** 2 **Source:** Native Locations **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/mobile/Library/Caches/com.apple.routined/Cache.sqlite-wal : 0x184977 (Table: ZRTCLLOCATIONMO, Size: 3485552 bytes)			

Figure 2.5.2 *Extracted GPS and native location entries showing consistent coordinates and time intervals of device activity.*

Correlated Wi-Fi Location Data

Several Harvested entries corresponded with Wi-Fi access points identified in the Wireless Networks dataset (see Section 2.4). Each access point BSSID was linked to coordinates

and timestamps, allowing potential triangulation of the device's position even in the absence of GPS data. This cross-correlation confirms that both Wi-Fi and GPS telemetry contributed to the device's geolocation record set, reinforcing the accuracy of movement reconstruction based on location-related artifacts.

Collectively, these findings demonstrate that the device maintained active location tracking functionality throughout the examined period, supporting other evidence of regular user activity.

2580		(40.737419, -73.877732)	**Type:** Harvested **Precision:** 177 **Name:** Wireless Networks Location - BSSId: 00:6C:BC:EF:19:8A **Time:** 9/29/2021 2:29:49 AM(UTC-4) **Source:** **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/root/Library/Caches/locationd/cache_encryptedB.db : 0x5255FB (Table: WifiLocation, Size: 44048384 bytes)	50	Wireless Networks	
2581		(40.736938, -73.878129)	**Type:** Harvested **Precision:** 37 **Name:** Wireless Networks Location - BSSId: 00:AC:E0:A5:16:C0 **Time:** 9/29/2021 2:29:49 AM(UTC-4) **Source:** **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/root/Library/Caches/locationd/cache_encryptedB.db : 0x5255BC (Table: WifiLocation, Size: 44048384 bytes)	50	Wireless Networks	
2582		(40.737297, -73.878548)	**Type:** Harvested **Precision:** 36 **Name:** Wireless Networks Location - BSSId: 00:EA:BD:FF:3C:41 **Time:** 9/29/2021 2:29:49 AM(UTC-4) **Source:** **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/root/Library/Caches/locationd/cache_encryptedB.db : 0x52557D (Table: WifiLocation, Size: 44048384 bytes)	50	Wireless Networks	
2583		(40.737129, -73.878731)	**Type:** Harvested **Precision:** 47 **Name:** Wireless Networks Location - BSSId: 10:D7:B0:E4:5F:AE **Time:** 9/29/2021 2:29:49 AM(UTC-4) **Source:** **Source file:** 271b2d62e4ab929f60e2ded6cfdb6e097f0f92d1_files_full.zip/private/var/root/Library/Caches/locationd/cache_encryptedB.db : 0x52553F (Table: WifiLocation, Size: 44048384 bytes)	50	Wireless Networks	
2584		(40.737777, -73.878373)	**Type:** Harvested **Precision:** 20 **Name:** Wireless Networks Location - BSSId: 14:59:C0:3D:5B:73	50	Wireless Networks	

Figure 2.5.3 *Wi-Fi–based harvested location entries with linked BSSIDs and positional data.*

2.6 Application Activity

Analysis of the applicationState.db and knowledgeC.db databases provided detailed insights into the device's installed applications, installation timestamps, and subsequent usage patterns. Both databases were parsed using Cellebrite Reader, and select tables were verified through DB Browser for SQLite to confirm record consistency.

Installed Applications

The applicationState.db contained records of both system and third-party applications installed on the device. Among these, several non-Apple applications were identified:

- bts.lunanueva
- com.cellhigh.MenopauseDiary2
- com.lbrc.PeriodCalendar
- com.menolabs.menolife
- com.womaneze.app

These applications are associated with health and menstrual tracking. Their collective presence indicates that the user engaged with multiple wellness-oriented applications within a short timeframe, suggesting focused activity around health monitoring.

DATA

	bundle_id	+
1	bts.lunanueva	
2	com.cellhigh.MenopauseDiary2	
3	com.lbrc.PeriodCalendar	
4	com.menolabs.menolife	
5	com.womaneze.app	
6		

Figure 2.6.1 *Extract showing non-Apple apps identified in the database (applicationState.db).*

Application Installations and First Usage

Entries within knowledgeC.db revealed installation timestamps for each application. All default Apple applications appeared to have been installed on 2021-09-12 at 03:48:00 UTC, aligning with initial device setup.

Subsequent installations of third-party apps occurred as follows:

- MenopauseDiary2 → Installed on 2021-09-18 at 23:53 UTC
- Lunanueva → Installed on 2021-09-22 at 21:17 UTC
- PeriodCalendar, Menolife, and Womaneze → Installed between 2021-09-22 21:16–21:20 UTC

These grouped timestamps suggest that multiple health-tracking applications were downloaded and configured in close succession, likely by the same user during a single session.

DATA

	day_utc	bundle_id	first_seen_install_utc	+
1	2021-09-12	com.apple.Bridge	2021-09-12 03:48:00	
2	2021-09-12	com.apple.DocumentsApp	2021-09-12 03:48:00	
3	2021-09-12	com.apple.Home	2021-09-12 03:48:00	
4	2021-09-12	com.apple.Maps	2021-09-12 03:48:00	
5	2021-09-12	com.apple.MobileAddressBook	2021-09-12 03:48:00	
6	2021-09-12	com.apple.MobileStore	2021-09-12 03:48:00	
7	2021-09-12	com.apple.Music	2021-09-12 03:48:00	
8	2021-09-12	com.apple.Translate	2021-09-12 03:48:00	
9	2021-09-12	com.apple.VoiceMemos	2021-09-12 03:48:00	
10	2021-09-12	com.apple.calculator	2021-09-12 03:48:00	
11	2021-09-12	com.apple.compass	2021-09-12 03:48:00	
12	2021-09-12	com.apple.facetime	2021-09-12 03:48:00	
13	2021-09-12	com.apple.iBooks	2021-09-12 03:48:00	
14	2021-09-12	com.apple.measure	2021-09-12 03:48:00	
15	2021-09-12	com.apple.mobilecal	2021-09-12 03:48:00	
16	2021-09-12	com.apple.mobilemail	2021-09-12 03:48:00	
17	2021-09-12	com.apple.mobilenotes	2021-09-12 03:48:00	
18	2021-09-12	com.apple.news	2021-09-12 03:48:00	
19	2021-09-12	com.apple.podcasts	2021-09-12 03:48:00	
20	2021-09-12	com.apple.reminders	2021-09-12 03:48:00	
21	2021-09-12	com.apple.shortcuts	2021-09-12 03:48:00	
22	2021-09-12	com.apple.stocks	2021-09-12 03:48:00	
23	2021-09-12	com.apple.tips	2021-09-12 03:48:00	
24	2021-09-12	com.apple.tv	2021-09-12 03:48:00	
25	2021-09-12	com.apple.weather	2021-09-12 03:48:00	
26	2021-09-18	com.cellhigh.MenopauseDiary2	2021-09-18 23:53:00	
27	2021-09-22	bts.lunanueva	2021-09-22 21:17:00	
28	2021-09-22	com.lbrc.PeriodCalendar	2021-09-22 21:16:00	
29	2021-09-22	com.menolabs.menolife	2021-09-22 20:42:00	
30	2021-09-22	com.womaneze.app	2021-09-22 21:16:00	

Figure 2.6.2 *Log of application installations and timestamps extracted from knowledgeC.db.*

Application Usage Summary

Analysis of the knowledgeC.db usage logs identified measurable activity for five non-system applications.

DATA

	bundle_id	sessions	total_seconds	total_minutes	+
1		1	53541.0	892.4	
2	com.cellhigh.MenopauseDiary2	19	2901.0	48.4	
3	com.apple.Preferences	27	1022.0	17.0	
4	com.menolabs.menolife	10	518.0	8.6	
5	com.apple.AppStore	14	398.0	6.6	
6	com.apple.purplebuddy	5	328.0	5.5	
7	com.apple.mobilemail	4	326.0	5.4	
8	com.apple.mobileslideshow	7	321.0	5.3	
9	bts.lunanueva	4	271.0	4.5	
10	com.apple.HealthENBuddy	2	103.0	1.7	
11	com.apple.Sharing.AirDrop	6	88.0	1.5	
12	com.apple.AuthKitUIService	1	39.0	0.7	
13	com.apple.springboard.today-view	4	17.0	0.3	
14	com.apple.AuthKitUI.AKSecondFactorAlert	1	19.0	0.3	
15	com.apple.quicklook.extension.previewUI	1	3.0	0.1	
16	com.apple.carkit.DNDBuddy	2	8.0	0.1	
17	com.apple.ContactsUI.ContactsViewService	1	5.0	0.1	
18	com.apple.springboard.app-library	1	2.0	0.0	
19	com.apple.corelocation.CoreLocationTemporaryPreciseAuthPromptPlugin	1	2.0	0.0	
20	com.apple.AuthKitUI.AKLocationSignInAlert	1	1.0	0.0	
21					

Figure 2.6.3 *Application usage recorded in minutes extracted from knowledgeC.db.*

Notably, MenopauseDiary2 showed extensive usage (approximately 15 hours total), strongly suggesting it was the primary application used for recording personal data or tracking events. Other apps, including Lunanueva and Menolife, displayed brief but repeated usage, indicating user experimentation or concurrent use for comparative purposes.

Widget Refresh and Device Activity

Corroborative data from knowledgeC.db under /dad/widgetRefresh streams indicated that these applications were periodically refreshing widgets and interacting with the system UI during regular device operation. This confirms active installation and use, as opposed to residual or deleted application data.

Collectively, these findings show a consistent behavioral pattern centered on health and wellness applications, establishing both the user's engagement with specific third-party services and the device's active operation during the final weeks of recorded activity.

2.7 Summary of Findings

This section consolidates the major evidentiary findings from the investigation:

Device and Ownership

- The extracted dataset originated from an Apple iPhone 8 (serial FFMWR1NRC6F, IMEI 35670108761405) identified as "Berg's iPhone."
- A single Apple ID (bergcybersectesting@gmail.com) was associated with the device, linked to multiple Apple services (iCloud, iTunes, Find My Friends, Game Center).
- The presence of synchronized iCloud and location services confirmed the device was active and personalized to one user profile.

Communications

- Email analysis revealed a configured Gmail account (bergcybersectesting@gmail.com) containing 134 total messages (131 inbound, 3 outbound).
- Outgoing messages were sent to alexongkowijoyo91@ymail.com between September 18–19, 2021, confirming authenticated outbound SMTP activity.
- Inbound messages included verification emails from Facebook, Google, Apple, WomanEze, and ProtonVPN, confirming ongoing service use and logins.
- No SMS, iMessage, or third-party chat application data was recoverable from the extracted filesystem.

Network and Connectivity

- Over 3,600 wireless network entries were recorded, providing extensive Wi-Fi BSSID data with geolocation coordinates.
- Multiple Bluetooth connections were logged with randomized MAC addresses, consistent with iOS privacy protocols.
- Wi-Fi connect/disconnect events correlated with active user interactions (screen on, device unlock), confirming real-time device usage.

Device Location

- The image contained 5,790 total location records, including 1,860 deleted but recoverable entries.
- Both GPS and Wi-Fi–derived location data were identified in Core Location and locationd caches.
- Coordinates clustered consistently around 40.73°N, 73.87°W, corresponding to a single geographic region of user activity during September 23 – October 1, 2021.

Applications

- The device contained multiple health-tracking applications, including MenopauseDiary2, Lunanueva, Menolife, PeriodCalendar, and Womaneze.
- Installation timestamps between September 18–22, 2021 indicated that these were downloaded and configured in close succession by the same user.
- Usage metrics showed MenopauseDiary2 as the most active app (≈15 hours total), indicating it was likely the primary wellness tracker.
- Widget refresh events in knowledgeC.db confirmed ongoing background activity, validating genuine app engagement.

Conclusion

The forensic analysis of the Apple iOS Full File System (UFDR) extraction provided a comprehensive overview of the device's operational state, user configuration, and behavioral patterns during late September 2021. The evidence demonstrates that the device was actively used by a single, identifiable user "Berg Cybersec", associated with the Gmail and Apple ID bergcybersectesting@gmail.com.

While no conventional messaging or call artifacts were present, the combination of email communications, network connectivity, location telemetry, and application activity establishes a clear timeline of active use. The device maintained consistent Wi-Fi and Bluetooth connectivity, regular iCloud synchronization, and continuous logging of geolocation data.

The user's focus on health-tracking applications and the installation of privacy-related tools (ProtonVPN) indicate both personal and security-conscious behaviors. These artifacts collectively portray an individual actively managing personal data and interacting across multiple online platforms during the extraction period.

No evidence of secondary user accounts, unauthorized access, or tampering was detected. All findings are derived directly from verified forensic artifacts, supported by validated hash integrity and tool-level verification within Cellebrite Reader.

This examination successfully documented device ownership, reconstructed user behavior patterns, and produced a complete artifact record of network, communication, and application activity. All findings are derived from verified forensic artifacts and supported by hash-validated evidence integrity maintained throughout the examination.